

Software Requirements Specification (SRS)

Document for Private Security Agency Licensing and HRMS

This SRS is organized per IEEE/ANSI 830-1993 standards, covering six main chapters: Introduction, Overall Description, Functional Requirements, External Interface Requirements, System Attributes (nonfunctional), and System Models (diagrams). It defines all requirements for a web-based licensing portal and embedded HRMS used by private security agencies.

Key goals include automating agency license applications and renewals, enforcing prerequisites, integrating payments (TeleBirr/banks), tracking agency location via GPS, integrating with national ID or Fayda and managing internal employee data. The document targets developers, testers, and stakeholders, detailing what the system will do and under what constraints.

Chapter 1: Introduction

1.1 Purpose

The project involves two interrelated systems: (1) a public-facing web portal for security facility licensing, and (2) an HRMS application for private security agencies. The portal will let citizens apply for new, renew, and track licenses online, access regulatory information, FAQs, and contact government offices, improving transparency and efficiency. The HRMS will help agencies manage personnel—tracking employee info, training, assignments, certifications, performance reviews, and compliance documents. Together, these systems replace slow manual processes with digital workflows, enhancing service delivery and data management.

Key objectives include easy online license applications with digital forms and reminders, an intuitive user interface (UI) with modern design, and a secure back-end for administrators. The HRMS will offer centralized document storage, automated reporting,

and role-based access. Both systems emphasize security, reliability, and maintainability. By automating workflows (e.g. license approval and employee onboarding), we expect faster processing and better data accuracy.

1.2 Scope

In Scope:

The software comprises two integrated components:

- **License Website:** Agencies use the portal to view General License Requirements (company registration, insurance, manager credentials, etc.). Features include online license application and renewal forms, downloadable documents, FAQ/feedback section, news/announcements display, contact information (office addresses, phones), and automated renewal reminders. It will also support admin interfaces for managing submissions and content. The admin side provides Federal Police users with a dashboard to manage agencies and applications. Optionally, a companion mobile app or responsive design may be included for mobile access.
- **HRMS Application:** An intranet or cloud-based application for Private Security Agencies. Core functions cover employee profiles, training and certification tracking, duty assignment (“protection program”) management, performance reviews, and document management (contracts, IDs, compliance documents). It will generate reports (e.g. headcount statistics, training completion) and support regulatory compliance (e.g. storing “certificate of competency” documents as per legal requirements). Both systems will include role-based access (e.g., public user, agency employee, administrators).

Out of Scope:

- Payroll, benefit management or scheduling beyond security assignments.
- Physical security devices (focus is software only).
- Integration with unrelated systems (e.g., external criminal databases unless explicitly required later).

The SRS does *not* assume existing legacy systems; if needed, integration is an assumption. It covers both software and high-level hardware/interface requirements but not detailed code or deployment.

1.3 Definitions, Acronyms, Abbreviations

SRS: Software Requirements Specification (this document).

API: Application Programming Interface (external service interface).

Admin User: Federal Police staff with management privileges.

Agency User: Authorized representative of a private security company.

HRMS: Human Resource Management System.

GPS: Global Positioning System (used to verify agency location).

TeleBirr: Ethiopia's mobile payment platform.

UI: User Interface.

Fayda: Ethiopia's national digital identification program.

DFD: Data Flow Diagram.

ERD: Entity-Relationship Diagram.

1.4 References

- IEEE Std 830-1998, *Recommended Practice for Software Requirements Specifications* (SRS) – standard structure and content.
- B. Penzenstadler et al., "Writing Requirements" (ODU CS course notes) – guidelines on separating functional vs. non-functional requirements.
- Wikipedia, *Software requirements specification* – discusses requirement quality (completeness, consistency, etc.).
- BMC Software Blog on SRS – outlines section organization (introduction, overall description, specific requirements) and data dictionary concept.
- Proc. No. 720/2011 Federal Police Commission Establishment Proclamation Grants FPC power to issue certificates to private agencies.
- Proc. No. 1284/2023 Ethiopian Digital Identification Proclamation Legal basis for mandatory Fayda integration.
- Proc. No. 714/2011 Public Servants' Pension Proclamation Governs social security registration for security personnel.
- Guideline 01/201 Federal Police Working Guideline for PSCs Defines training syllabus and guard obligations.

1.5 Document Overview

This document follows a six-chapter IEEE SRS structure: Chapter 1 (this section) sets context. Chapter 2 provides an overall description of the system (perspective, functions, user classes, environment, constraints, assumptions). Chapter 3 lists detailed functional requirements by feature. Chapter 4 covers external interfaces (UI, hardware, software, communication). Chapter 5 enumerates non-functional requirements (performance, security, etc.). Chapter 6 presents system models (use cases, diagrams, tables). Table of Contents, lists of figures/tables, and a glossary/index are included.

Chapter 2: Overall Description

The system is designed to act as a centralized authority for the private security sector, which has seen massive growth over the last three decades in Ethiopia. From an initial period in the 1990s characterized by demobilized soldiers to the intensification period of 2008-2009, the industry now employs thousands of guards across over 100 companies in Addis Ababa alone. The transition to a digital system is an essential response to the rising security threats and the need for standardized vetting.

2.1 Product Perspective

These are new systems; they stand alone but may interface with existing government networks or authentication services. The License Website will be an Internet-facing portal, implemented perhaps on a government web server or cloud. It may integrate with the Department's internal database for license records (if one exists). The HRMS is for use by private security agencies and may run on a secure cloud platform or agency servers. Both should use standard web architecture (e.g. client-server).

A conceptual architecture:

- Frontend: Web/Mobile user interface (UI) for applicants and employees.
- Backend: Application server (business logic) and database.

- Interfaces: Email/SMS gateway for notifications; possibly integration with Fayda ID verification or mobile app notification services.

Constraints include regulatory compliance (Ethiopian security laws and data protection rules) and any ICT infrastructure policies of the agencies.

2.2 Product Functions

2.2.1 External Agency Portal Functions

The portal serves as the primary interface for agencies to interact with the regulator. Its core function is the management of the "Certificate of Competency." Agencies can browse the general requirements for licensing on the homepage before logging in. Once authenticated via Fayda, they can initiate new applications or biennial renewals. A significant function of the portal is the restriction of data changes; specifically, agencies are prohibited from altering their headquarters' coordinates or their designated manager without triggering a formal notification and approval workflow with the Federal Police.

2.2.2 Administrative Dashboard Functions

The Federal Police Admin dashboard provides a high-level view of the industry. It includes tools for managing agency profiles, reviewing licensing applications, and making decisions (Approve, Pending, or Reject). The dashboard also includes a "Notification Button" for broadcasting new regulations or emergency directives via SMS and email. The reporting module allows admins to generate statistics on guard training levels, gender distribution, and geographic density.

2.2.3 Internal Human Resource Management (HRMS)

The internal HRMS is a mission-critical module that allows agencies to manage their employees while providing the Federal Police with visibility into the workforce. It tracks personal profiles, work history, training certifications (30-day programs), and performance reviews. This ensures that guards who are fired for disciplinary reasons in one agency can be identified if they attempt to join another.

2.3 User Classes and Characteristics

- Guest/General User: Can view publicly available info (prerequisites, FAQs) but cannot apply without an account or sign up.
- System Administrator (Federal Police): Highly trained user. Manages all agencies and applications via admin interface and staff who update website content (news, FAQs), respond to feedback, and post announcements.
- Security Agency Employees: Guard staff who may use HRMS to view their own profile or training. Must be able to update personal info and access assignment schedules.
- HRMS Administrators/Managers: Security agency HR or management staff. They manage employee records, create assignments, and run reports. Moderate technical skill.
- System Administrators: IT personnel for Department or Agency. They have privileges to configure the system, manage user accounts, and perform backups.

2.4 Operating Environment

- Web Platform: Modern web browsers (Chrome, Firefox, Edge, Safari) on desktop and mobile. The website should be responsive.
- Server: Can run on Linux or Windows servers. We assume a typical 64-bit environment, possibly cloud-based (e.g. AWS, Azure) or government datacenter.
- Database: Relational DBMS (e.g. MySQL, PostgreSQL, or SQL Server).
- Hardware: Standard web server, database server. Possibly separate servers for production vs. testing. Minimum specs assumed are a multi-core CPU and 8+ GB RAM.
- Mobile App (optional): If implemented, Android/iOS platforms. Otherwise, mobile web design must be considered.
- Software Stack: For example, Node.js or Java Spring for backend; React or Vue for frontend; PostgreSQL or MySQL for DB. (Choice discussed below.)
- Network: Internet connectivity required. HTTPS (TLS) for secure communication.

→ Services: TeleBirr and bank payment systems via their provided APIs (assuming connectivity), SMS gateway and email SMTP for notifications.

2.5 Design and Implementation Constraints

The system is bounded by several critical constraints that ensure it remains a tool for law enforcement and not just a business management app:

1. **Legal Exclusivity:** The system must only allow registrations for private security agencies. Any attempt to register as a public person or for a firearms license must be programmatically blocked.
2. **Location/Manager Locking:** In accordance with Federal Police requirements, the system must "lock" the bureau location and manager fields. Any change requires a formal notification to the admin, and the system must use GPS to verify the new location before approval.
3. **National ID Sovereignty:** Login and identity verification must use the Fayda platform. The system cannot maintain its own independent identity registry for high-level personnel.
4. **Financial Compliance:** Licenses can only be "Approved" for issuance after a successful payment transaction is verified by the Telebirr or banking gateway.
5. **Data Retention:** Employee fingerprint and warranty data must be retained as long as the guard is active in the industry to ensure compliance with Article 11 of the vocational counseling service division.

2.6 Assumptions and Dependencies

Assumes users have basic literacy and internet access.

TeleBirr/bank APIs remain available and have negligible downtime.

Mobile devices provide reliable GPS location.

Government regulations (requirements, fees) are provided externally; the system will need updates if rules change.

Integration with an external HR management system is not required. This HRMS is separate.

Localization: System is initially English-only. (Future multilingual support is out of scope unless specified later.)

Concurrency: Assuming up to 200 concurrent users (agencies+admins). This is an assumption to be validated

2.7 Technology Stack Options (Comparison)

We compare plausible tech stacks below:

Component	Option A: LAMP Stack	Option B: MERN Stack	Option C: Python/Django + React	Note
Backend	PHP (Laravel) + MySQL	Node.js/ Express + MongoDB	Python/Django + PostgreSQL	Laravel/Django are Mature frameworks; Node/Mongo is scalable and JSON-friendly
Frontend	Server-rendered HTML/CSS	React or Angular SPA	React (SPA)	React works with any backend; Angular is more opinionated.
Hosting	Shared Linux hosting (cheap)	Cloud service (AWS/Azure)	Cloud or on prem (flexible)	Cloud offers easy scaling; shared hosting is limited for high traffic.
Security	Well-known LAMP vulnerabilities	Modern token auth (JWT)	Strong libraries (OAuth, JWT)	All have security best practices; e.g. PHP/ Node need careful patching.
Maintainability	Large community (PHP)	JavaScript everywhere	Python's readability	Team's expertise will guide choice.
Performance	Decent for	High	Good with	Node.js handles

	moderate load	performance at scale	optimizations	many concurrent requests well; PHP is stable for smaller loads.
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Table 1: *Technology stack options with popular choices*. Each has pros/cons. *Node.js* offers non-blocking I/O and a unified JS codebase; *Django* provides batteries-included security features; *Laravel* is easy to host. On the front end, *React* has a large ecosystem, *Angular* is opinionated/enterprise-ready, *Vue* is lightweight. *PostgreSQL/MySQL* are robust SQL options; *MongoDB* suits flexible schemas. *AWS* and *Azure* provide scalable cloud hosting, while *on-premises* may satisfy local data control requirements. The final choice will depend on the team's expertise and infrastructure preferences (to be decided by stakeholders).

Chapter 3: System Features (Functional Requirements)

This chapter breaks down the functional requirements of the system into granular, testable specifications, focusing on the external interfaces and core system features requested by the Federal Police.

3.1 External Interface Requirements

The system must present a seamless interface to users, hardware, and external software ecosystems.

3.1.1 User Interfaces (UI)

The UI is designed to be "visually appealing and user-friendly," accommodating both desktop and mobile applications.

- **Public Face Homepage:** This page must be accessible without login. It displays the comprehensive list of "Requirements to have a license," including brand

name designation, business pre-registration, TIN, articles of association, bylaws, insurance, and bank statements.

- User Login/Dashboard: After authenticating via Fayda, the user sees a dashboard showing their current license status, any "Renewal Notifications," and links to "New Application" or "Renewal" forms.
- Admin Command Center: The admin interface is centered around a dashboard with real-time analytics. It includes pages for "Private Security Agencies Management," "License Management," "New Application Review," and "Reports".

3.1.2 Hardware Interfaces

The system interfaces with GPS-enabled devices to monitor office location compliance. For agencies, this may involve dedicated GPS trackers installed at the bureau or the use of mobile apps that report "Latitude/Longitude" coordinates during clock-in and reporting activities. The system must be capable of receiving coordinates with an accuracy radius of 50 meters or less to minimize "false alerts" caused by GPS bounce.

3.1.3 Software Interfaces

A critical component is the integration with national digital services.

Software Interface	Protocol	Description
Fayda OIDC	OpenID Connect / OAuth 2.0	Brokering identity verification for agency managers and owners.
Telebirr C2B	REST API / H5	Handling the payment of license fees and generating transaction IDs.

SMS/Email Gateway	REST / SMTP	Automated delivery of "Renewal Reminders" and "New Regulation" notifications.
Bank API (Chapa/AddisPay)	Webhooks / REST	Verifying high-value bank transfers for large agency licenses.

3.1.4 Communications Interfaces

All communication between clients and servers uses HTTPS (SSL/TLS). Payment and SMS APIs also use HTTPS on standard ports. The portal's web API endpoints will follow REST principles. No special protocols (like WebSockets) are needed. The system assumes reliable internet connectivity; offline mode is *not* supported in version 1.

3.2 System Features

The system's core logic is organized into three major functional domains: Licensing Management, Compliance & Notifications, and the Internal HRMS.

3.2.1 Home/Info Page

Shows general licensing rules and requirements, announcements, FAQ, services, download forms and guidelines, contact information and remainders notification.

- Announcements/News: Admins can post announcements or news (e.g. changes in regulations). The homepage or a news section will display these updates in reverse-chronological order.
- Download Forms/Guidelines: Provide a "Downloads" page with PDF forms and guidelines for license and security facility applications.
- FAQ and Feedback: A page to list Frequently Asked Questions. Staff can update this list. Users can submit feedback or questions via a web form.

- Contact Information: Display the Department's office address, phone, fax, email, and working hours. Possibly an embedded map. (From PDF: *"Address, phone number, or address that tells where to go if you need to come to the office"*)
- User Registration & Login: Users must register with basic identity info (name, email, password) to use online services. (If no login is needed, the form can include contact info.) (See Use Case *Register/Login*). Provide password reset via email.
- Services: The system provides multiple services for private security agencies including license requirement viewing, new license application submission, license renewal, payment processing via Telebirr and banks, notification services via email and SMS, agency profile management with GPS-based restrictions, employee human resource management, and application status tracking. These services are accessible through a user-friendly web interface and are designed to ensure compliance with Federal Police regulations.

3.2.2 New Application and Licensing Workflow

The system facilitates a "transparent and efficient" way for agencies to apply for their initial Certificate of Competency.

- Document Upload: The system shall provide an interface for agencies to upload a photo of digital assets, including the uniform sample, logo, and employee ID samples.
- Management Profile: Applicants must provide biographical data for the Manager, Head of Operations, and Head of Administration. This includes fingerprints, medical results, proof of education, and evidence of prior experience in the Defense Forces or Police.
- Vetting Queue: Once submitted, the application enters the "New Application Management Page" on the admin side. The admin can view all 18+ required documents and decide to "Approve," "Pending," or "Reject" the application.

- **Payment Trigger:** Upon admin approval, the system triggers a payment prompt. The agency must pay the license fee via Telebirr or integrated banks before the final digital certificate is issued.

3.2.3 Renewal and Compliance Tracking

Licenses must be renewed every one year, a process the system automates to ensure "licenses are always up-to-date".

- **Automated Reminders:** The system shall send automatic reminders via SMS and email 30 days before the license expiration.
- **Change Restriction Logic:** The system must block any change to the "Bureau Location" or "Company Manager" during the renewal process unless a formal notification is sent to the Federal Police admin.
- **GPS Verification:** If a change of location is requested, the system must trigger a GPS verification. The agency must provide the new coordinates, which are then verified by a technical committee during a physical inspection.

3.2.4 Internal HRMS for Security Agencies

This module enables the Federal Police to "monitor security service providers" and control their human resources.

- ❖ **Employee Profiles:** A complete repository for personal addresses, work experience, certificates, and work history.
- ❖ **Training Tracker:** Specifically tracks whether guards have received 15-day 30-day training programs and verifies their certificates of competency.
- ❖ **Military Intelligence Reporting:** A tool for agencies to submit "daily, monthly, quarterly, and annual military intelligence reports" regarding security and data privacy.
- ❖ **Duty Assignments:** Agencies can define protection details and assign employees to shifts. The system tracks who is on duty where, and records attendance.
- ❖ **Disciplinary & Performance:** A performance review report module that logs disciplinary actions and commendations to help the Federal Police make informed decisions about industry efficiency.

- ❖ Document Repository: A central storage for agency documents (e.g. company bylaws, insurance, bank statements) and employee documents (IDs, contracts). Documents are linked to the relevant agency or person.

3.2.5 Administrative Controls

The admin interface provides the tools for high-level oversight.

- Dashboard Analytics: Real-time visualization of the number of agencies, active licenses, and pending applications.
- Private Security Agencies Management: A page for viewing the history of every agency, including their previous locations and managers.
- License Application Management (Admin): Admins process each new or renewal application. For each submission, admins see all entered data and uploaded files. The admin interface allows setting the application status to *Approved*, *Pending*, or *Rejected*. If *Approved*, an invoice is generated and sent to the agency. If *Rejected*, the admin provides a reason which is sent to the applicant.
- Settings Page: Allows admins to modify the global requirements displayed on the homepage and update the automated email/SMS templates.
- Notification Button: A broadcast tool for sending urgent "New Regulation" messages to all registered agencies at once.
- Reporting (Admin): The system can generate custom reports. For example, admins can request a report of "All licenses issued in the last year" or "All agencies with expiring licenses next month". Reports can be exported in PDF or Excel. Key metrics like total fees collected, number of active agencies, or average processing time should be included.

Chapter 4: Non-Functional Requirements

This section describes system-wide qualities and constraints.

Performance: The system must handle moderate load (e.g. up to 200 concurrent agency users). Interactive operations (form submission, search) should complete quickly (ideally under 0.5 seconds). Bulk report generation (for large agencies) should

finish within 10 seconds. Backend processes (approval workflows) should execute within a few seconds to keep the UI responsive.

Security: Access control is mandatory. Agency users can only see their own data; admin users have broader privileges. All pages and APIs must require login (sessions should expire after inactivity). Sensitive data (passwords, personal details) must be encrypted. The system should protect against common threats (SQL injection, XSS). Audit logs will record all login attempts and actions like application approvals. Administrators should use strong, unique credentials (multi-factor authentication for admin accounts is recommended).

Reliability & Availability: The system should be up 99.5% of the time (approximately <4 hours downtime per month). Database and code will be backed up daily. In case of a crash, the latest backup should restore all recent data with minimal loss. The system must fail gracefully: if a payment or SMS service is down, the user should see a clear error message rather than a crash.

Maintainability: Code must be modular and well-documented. Requirements will be tracked in a data dictionary to avoid confusion. Clear error logging is required. Updates (e.g. adding a new payment partner) should be possible without major rewrites.

Portability: The application shall be platform-independent. It should run on either Windows or Linux servers. The UI uses standard HTML/CSS/JavaScript so it works across common web browsers. We avoid OS-specific assumptions.

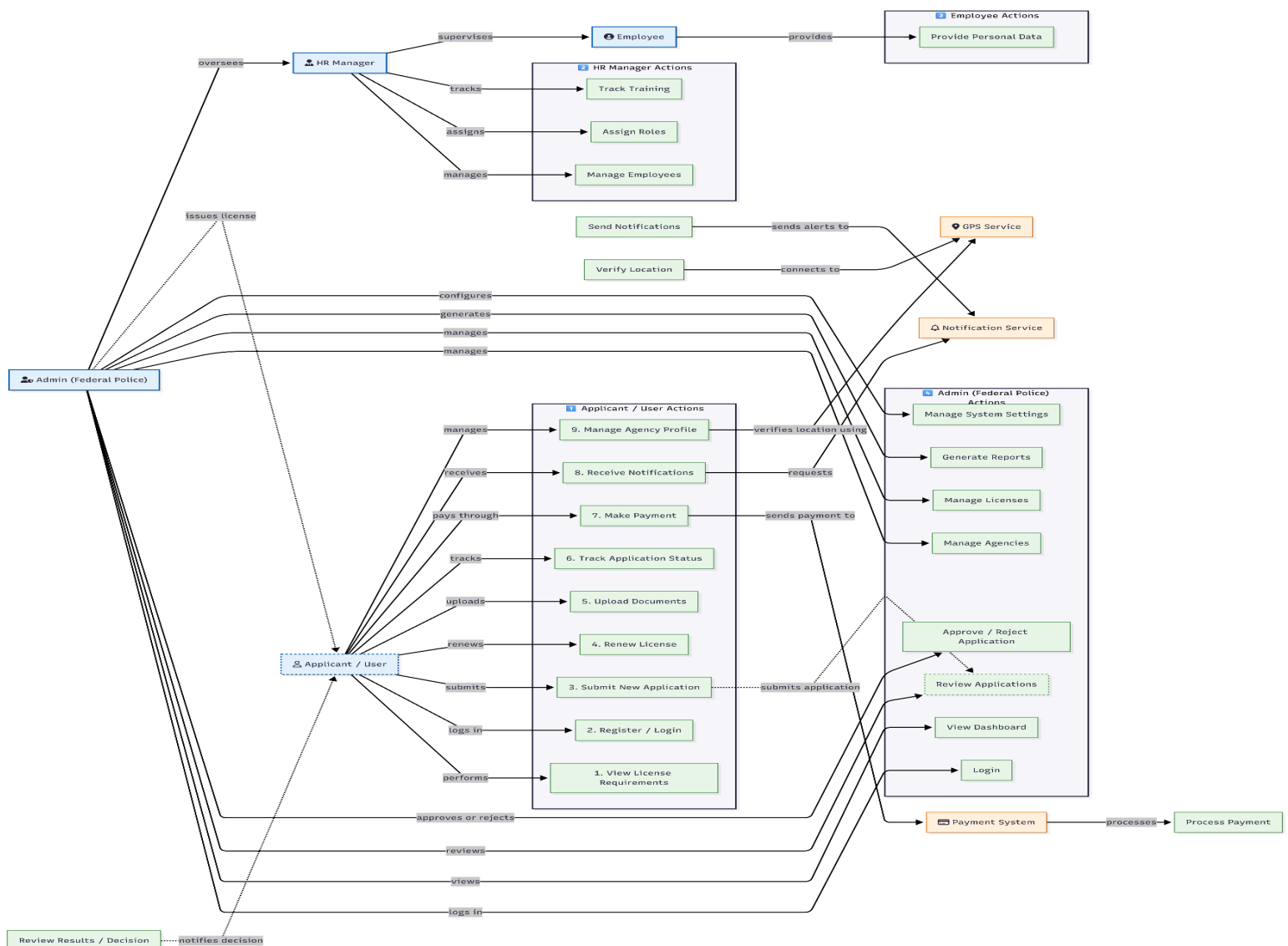
Scalability: The architecture should allow adding servers as load grows (e.g. moving from single to multiple web or database servers). Horizontal scaling (load balancing) should be possible.

Data Quality: All input data must be validated. For example, the license fee must be numeric within allowed ranges. Each requirement is complete, consistent, and unambiguous by design.

Chapter 5: System Models and Analysis

5.1 Use Cases Diagram and Description

Key use cases are described below, with actors and main flows. A UML use case diagram illustrates their relationships:



UC-01: View License Requirements

Primary Actor: Applicant / User

Secondary Actor: System

Description:

This use case allows applicants to view all requirements necessary to obtain a private security agency license before initiating the application process.

Preconditions:- The system must be operational and accessible.

Main Flow:

1. The user accesses the system homepage.
2. The user navigates to the “License Requirements” section.
3. The system displays all required documents, eligibility criteria, and legal conditions.

Postconditions:- The user is informed about all necessary requirements.

Priority: High

UC-02: Register / Login

Primary Actor: Applicant / User, Admin

Description:

This use case enables users and administrators to securely access the system using valid credentials.

Preconditions:

- The user must have a registered account.

Main Flow:

1. The user enters username and password.
2. The system validates the credentials.
3. The system grants access to the appropriate dashboard.

Alternative Flow:

- If credentials are invalid, the system displays an error message.

Postconditions:

- The user is authenticated and granted access.

Priority: High

UC-03: Request to Start Application

Primary Actor: Applicant / User

Secondary Actors: Admin, Notification Service

Description:

This use case allows an applicant to request permission to initiate a license application process.

Preconditions:

- The applicant must be logged into the system.

Main Flow:

1. The applicant selects “Request to Start Application.”
2. The system records the request.
3. The system sends notification (SMS and Email) to the Admin.

Postconditions:

- The Admin is notified of the request.

Priority: High

UC-04: Submit New Application

Primary Actor: Applicant / User

Secondary Actors: System, Notification Service

Description:

This use case enables applicants to submit a new license application.

Preconditions:

- The applicant must have fulfilled all requirements.
- The applicant must have permission to apply.

Main Flow:

1. The applicant selects “New Application.”
2. The applicant fills out the application form.
3. The applicant uploads required documents.
4. The applicant submits the application.
5. The system validates all inputs.
6. The system stores the application with a “Pending” status.

7. The system notifies the Admin via SMS and Email.

Postconditions:

- The application is successfully recorded in the system.

Priority: High

UC-05: Renew License

Primary Actor: Applicant / User

Secondary Actors: System, Notification Service

Description:

This use case allows applicants to renew an existing license.

Preconditions:

- The applicant must have an existing license.

Main Flow:

1. The applicant selects the license to renew.
2. The applicant updates required information.
3. The applicant uploads updated documents.
4. The applicant submits the renewal request.
5. The system records the request.
6. The system notifies the Admin.

Postconditions:

- The renewal request is stored with a pending status.

Priority: High

UC-06: Upload Documents

Primary Actor: Applicant / User

Description:

This use case allows applicants to upload required documents for applications or renewals.

Preconditions:

- The applicant must be logged in.

Main Flow:

1. The applicant selects a file.
2. The system validates file type and size.
3. The system uploads and stores the document securely.

Postconditions:

- The document is successfully stored in the system.

Priority: High

UC-07: Track Application Status

Primary Actor: Applicant / User

Description:

This use case allows applicants to track the progress of their applications.

Preconditions:

- The applicant must have submitted an application.

Main Flow:

1. The applicant accesses the dashboard.
2. The system displays application status (Pending, Approved, Rejected).

Postconditions:

- The applicant is informed of the application status.

Priority: High

UC-08: Make Payment

Primary Actor: Applicant / User

Secondary Actor: Payment System (Telebirr / Bank APIs)

Description:

This use case allows applicants to pay license fees after approval.

Preconditions:

- The application must be approved.

Main Flow:

1. The system displays payment options.
2. The applicant selects a payment method (Telebirr or Bank).
3. The payment system processes the transaction.

4. The system receives payment confirmation.
5. The system activates the license.

Postconditions:

- Payment is completed and license is activated.

Priority: High

UC-09: Notification Management

Primary Actor: System

Secondary Actors: Applicant / User, Admin, Notification Service

Description:

This use case handles all system-generated notifications sent via SMS and Email.

Preconditions:

- Notification service must be active.

Main Flow:

1. The system monitors system events.
2. When a trigger occurs, the system sends notifications via SMS and Email.

Notification Events:

- License expiry reminder (30 days before expiration) sent to Applicant
- New application request notification sent to Admin
- New application submission notification sent to Admin
- Renewal application submission notification sent to Admin
- Application approval/rejection notification sent to Applicant

Postconditions:

- Relevant users receive timely notifications.

Priority: High

UC-10: Manage Agency Profile

Primary Actor: Applicant / User

Secondary Actors: Admin, GPS Service

Description:

This use case allows applicants to manage agency details with strict validation rules.

Preconditions:

- The applicant must be logged in.

Main Flow:

1. The applicant updates profile information.
2. If location is changed, the GPS system verifies the location.
3. If the manager or location is changed, the system sends a request to Admin.
4. Admin reviews and approves/rejects the change.

Postconditions:

- Profile is updated only after validation and approval.

Priority: High

UC-11: Manage Employees (HR System)

Primary Actor: HR Manager

Secondary Actor: Employee

Description:

This use case enables HR managers to manage employee information within the agency.

Preconditions:

- HR Manager must be authenticated.

Main Flow:

1. HR Manager adds or edits employee details.
2. HR Manager assigns roles and responsibilities.
3. HR Manager tracks training and performance records.
4. The system stores the information.

Postconditions:

- Employee records are updated.

Priority: Medium

UC-12: Provide Employee Data

Primary Actor: Employee

Description:

This use case allows employees to provide personal and job-related information.

Preconditions:

- Employees must be registered in the system.

Main Flow:

1. Employee submits personal data.
2. The system stores the information.

Postconditions:

- Employee data is recorded.

Priority: Medium

UC-13: Review Applications

Primary Actor: Admin

Description:

This use case allows the Admin to review submitted applications.

Preconditions:

- Applications must exist in the system.

Main Flow:

1. Admin views pending applications.
2. Admin reviews submitted documents.

Postconditions:

- Applications are ready for decision-making.

Priority: High

UC-14: Approve / Reject Application

Primary Actor: Admin

Secondary Actor: Notification Service

Description:

This use case allows Admin to approve or reject applications.

Preconditions:

- Application must be reviewed.

Main Flow:

1. Admin selects an application.
2. Admin chooses to approve or reject.
3. The system updates the status.
4. The system sends notification to the Applicant.

Postconditions:

- Application status is updated.

Priority: High

UC-15: Manage Agencies

Primary Actor: Admin

Description:

This use case allows Admin to manage and monitor private security agencies.

Main Flow:

1. Admin views agency list.
2. Admin monitors changes and compliance.

Postconditions:

- Agencies are properly managed.

Priority: High

UC-16: Generate Reports

Primary Actor: Admin

Description:

This use case allows Admin to generate various reports.

Main Flow:

1. Admin selects report type.
2. The system generates reports (license, payment, HR).

Postconditions:

- Reports are generated successfully.

Priority: Medium

UC-17: Manage System Settings

Primary Actor: Admin

Description:

This use case allows Admin to configure system rules and parameters.

Main Flow:

1. Admin updates system settings (fees, rules, regulations).
2. The system saves changes.

Postconditions:

- System configuration is updated.

Priority: Medium

UC-18: GPS Location Verification

Primary Actor: GPS Service

Description:

This use case ensures that agency location changes are verified.

Main Flow:

1. The system requests location verification.
2. GPS service validates location.
3. System processes result.

Postconditions:

- Location changes are validated.

Priority: High

UC-19: Process Payment

Primary Actor: Payment System

Description:

This use case handles financial transactions securely.

Main Flow:

1. The system sends payment requests.
2. The payment system processes transactions.
3. Payment system returns confirmation.

Postconditions:

- Payment status is updated.

Priority: High

UC-20: Send Notifications

Primary Actor: Notification Service

Description:

This use case handles delivery of SMS and Email messages.

Main Flow:

1. The system sends notification requests.
2. Notification service delivers messages.

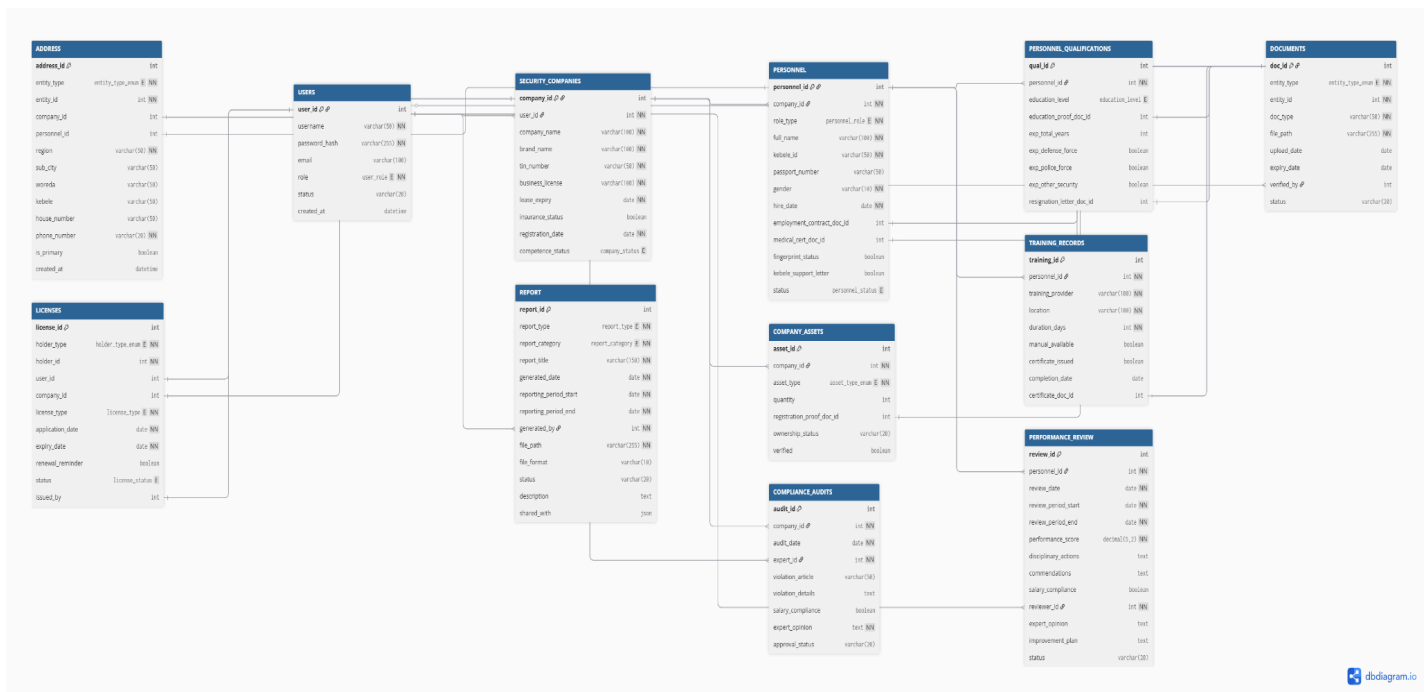
Postconditions:

- Messages are successfully delivered.

Priority: High

5.2 Entity-Relationship Diagram (ERD)

The ERD defines the database schema. Core entities and relationships include:



5.3 DATA FLOW DIAGRAM (DFD)

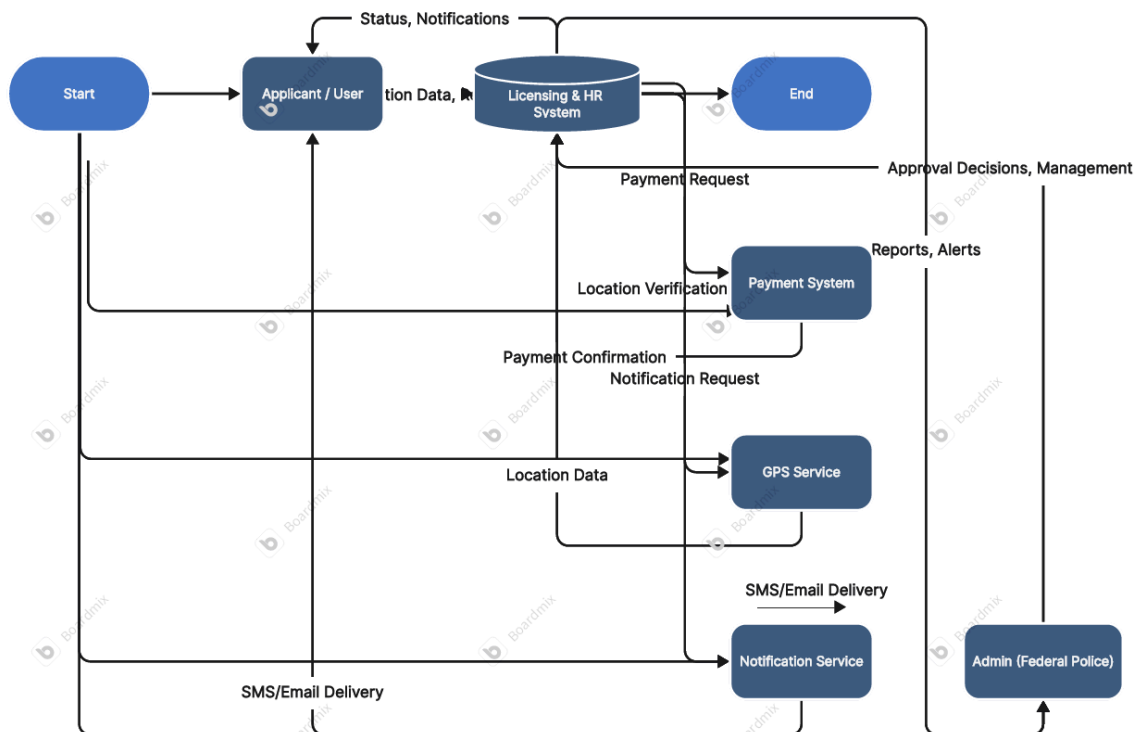
5.3.1 Context Diagram (Level 0 DFD)

The Context Diagram represents the entire system as a single process and shows how it interacts with external entities such as users, admin, and external services.

Actors (External Entities):

- Applicant / User
- Admin (Federal Police)
- Payment System (Telebirr / Bank APIs)
- Notification Service (SMS / Email Gateway)
- GPS Service (Location Verification System)

Process:- Private Security Licensing & HR Management System



6.3.2 Level 1 Data Flow Diagram

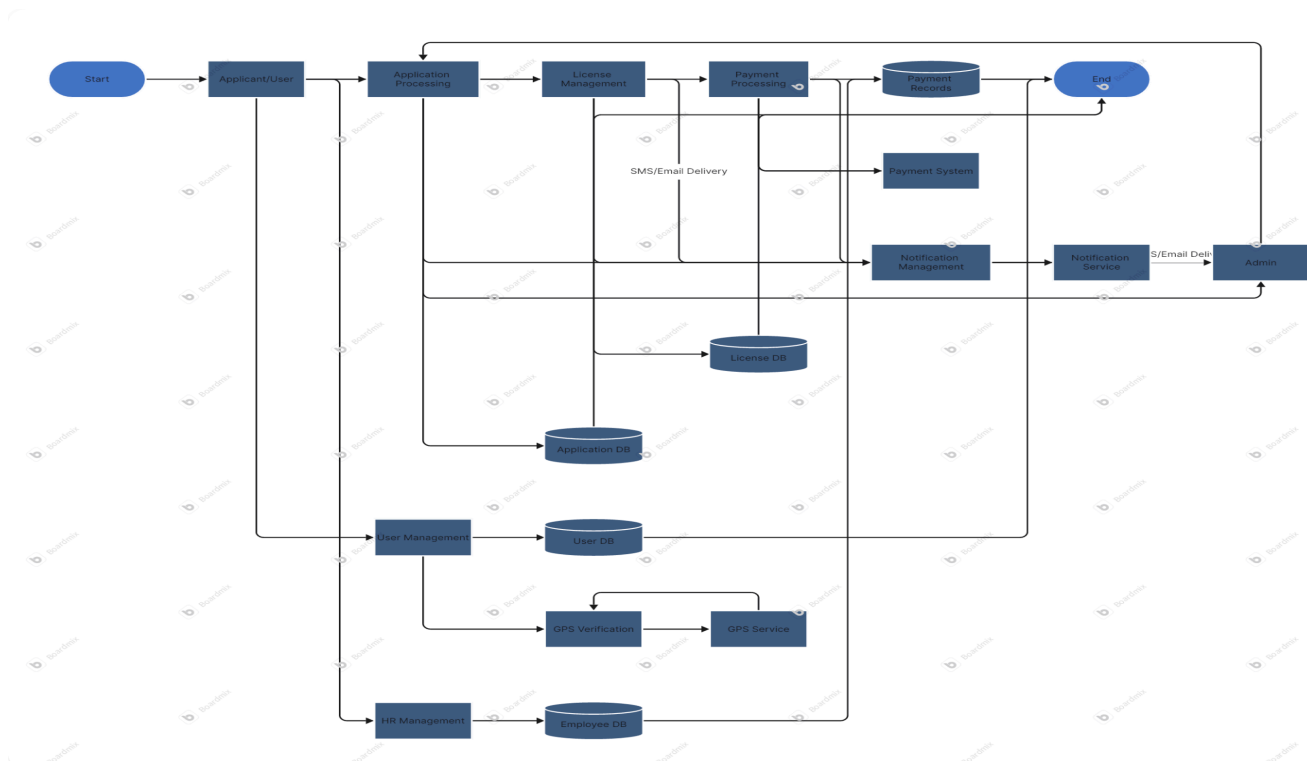
This diagram breaks down the system into major processes and shows how data flows between them.

Main Processes:

1. User Management
2. Application Processing
3. License Management
4. Payment Processing
5. Notification Management
6. HR Management
7. GPS Verification

Data Stores:

- D1: User Database
- D2: Application Database
- D3: License Database
- D4: Employee Database
- D5: Payment Records



5.3.3 DFD Explanation

The Data Flow Diagram illustrates how data moves within the Private Security Licensing and Human Resource Management System.

At the highest level (Context Diagram), the system interacts with external entities such as Applicants, Admin, Payment Systems, Notification Services, and GPS Services.

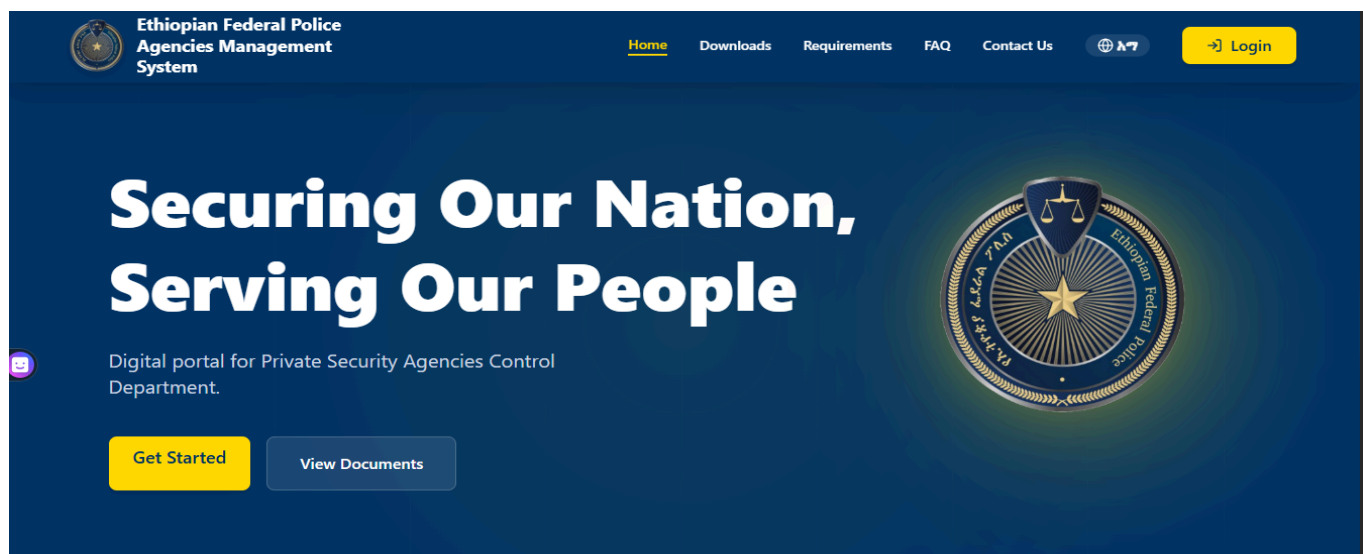
At Level 1, the system is decomposed into several key processes:

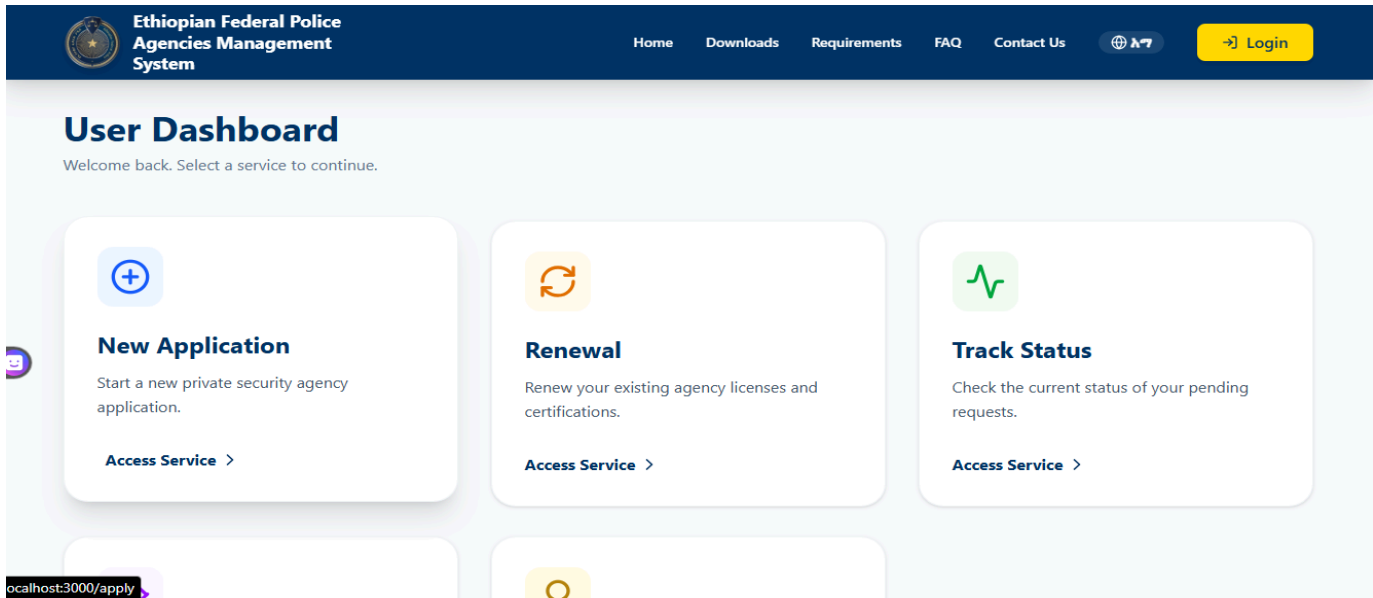
- User Management: Handles user registration, login, and profile updates.
- Application Processing: Manages submission and review of license applications and renewals.
- License Management: Handles license issuance and updates after approval.
- Payment Processing: Integrates with Telebirr and banking systems to process payments.
- Notification Management: Sends SMS and Email notifications for events such as application submission, approval, and renewal reminders (30 days before expiry).
- HR Management: Manages employee records within private security agencies.
- GPS Verification: Validates agency location changes to ensure compliance.

The system also interacts with multiple data stores to maintain persistent data including user information, applications, licenses, employees, and payment records.

5.4 UI Wireframes and Navigation

Key UI screens (wireframes) might include the License Application page, the HRMS dashboard, and a report view. For example, a conceptual wireframe could look like this:





Conceptual UI wireframe for the agency portal. This mockup sketch illustrates the user portal layout. The left panel (highlighted) shows the list of License Prerequisites (the “Requirements to Have a License”). The right panel includes login and application buttons. After login, the interface will similarly display forms for “New Application” and “Renewal,” along with any alerts (e.g. “Your license expires in 20 days”).

Chapter 6: Appendices and Index

6.1 Glossary

The following terms are frequently used throughout this Software Requirements Specification (SRS):

- **Agency:** A private security organization applying for and managing licenses.
- **Applicant/User:** A registered private security agency user interacting with the system.
- **Admin:** Federal Police administrator responsible for reviewing and managing applications.
- **HRMS (Human Resource Management System):** A subsystem used to manage employees of private security agencies.

- **License Application:** A request submitted by an agency to obtain or renew a license.
- **Telebirr:** A mobile payment service used for processing license payments.
- **GPS Service:** A system used to verify the physical location of agencies.
- **Notification Service:** A system responsible for sending SMS and Email notifications.
- **Payment System:** External systems such as Telebirr and banking APIs used to process transactions.
- **Renewal:** The process of extending the validity of an existing license.

6.2 Analysis of Ambiguities and Assumptions

Several items were not specified in the source document and are assumed or flagged for clarification:

Deployment Environment: It is unclear whether the systems should be hosted on-premises or in the cloud. *Assume:* A cloud deployment (e.g. AWS or Azure) for scalability, unless otherwise directed.

Target Platforms: The questionnaire mentions a mobile app as optional. If a native app is required, target OS versions should be clarified. *Assume:* At minimum, mobile web support (responsive design); Android/iOS apps are a future possibility.

User Roles: Exact user roles and hierarchy (especially for HRMS) are not fully detailed. We assume roles: Employee, Manager, HR Admin, System Admin. If “Supervisor” is a separate role, it should be defined.

Performance Targets: No metrics (e.g. number of users, response times) were given. *Assume:* Up to a few hundred concurrent users for reasonable load. This can be refined with performance testing.

Languages: No mention of multilingual support. *Assume:* Primary UI in English; localization is out of scope for now.

Legal Compliance: The specific legal requirements (e.g. retention period for records) are not detailed. *Assume:* Follow general regulations; adjust if the agency provides laws.

Integration with Other Systems: The requirements do not specify integration with government databases (e.g. national ID). *Assume:* Minimal external integration. This may need revision later.

Email/SMS Gateways: Choice of provider is unspecified. *Assume:* Use a reliable third-party API (like Twilio, SendGrid) depending on budget.

Technology Stack: No tech stack is mandated. The following table compares example options for core components.

7.3 Traceability Matrix (Outline)

For completeness, the final system shall include a traceability matrix linking requirements to their sources and corresponding test cases.

Requirement ID	Description	Source (Interview/Doc)	Related Use Case	Test Case ID
FR-01	User Registration	Stakeholder Interview	UC-02	TC-01
FR-02	Submit Application	Requirement Document	UC-04	TC-02
FR-03	Payment Processing	System Design	UC-08	TC-03
FR-04	Notification System	Stakeholder Need	UC-09	TC-04

7.4 Data Dictionary

This section defines key data elements used in the system.

Field Name	Description	Data Type
agencyID	Unique identifier for agency	Integer
userID	Unique identifier for user	Integer
applicationID	Unique ID for license application	Integer
licenseID	Unique ID for license	Integer
employeeID	Unique identifier for employee	Integer
paymentID	Unique identifier for payment transaction	Integer
status	Status of application (Pending/Approved/Rejected)	String
expiryDate	License expiration date	Date
gpsLocation	Verified location of agency	String

6.5 Index

The following index provides quick reference to important terms used in this document:

- Agency
- Applicant
- Admin
- Employee
- HRMS
- License Application
- Renewal

- Telebirr
- GPS
- Notification
- Payment System

6.6 Testable Requirements

Each functional requirement is stated so it can be verified by testing

. For example:

- *Requirement*: “The system shall allow uploading of PDF documents up to 5 MB.”

Test: Attempt to upload a 6 MB PDF (should be rejected) and a 3 MB PDF (should succeed).

- *Requirement*: “Expiry reminders shall be sent 30 days before license expiration.”

Test: Set a license expiry 30 days ahead and check that exactly one reminder email is sent on the correct day.

- *Requirement*: “HR admins can export employee data to CSV.”

Test: Click “Export” and verify the CSV contains all current employees.

Each requirement will be mapped to one or more test cases during validation.

Requirements should avoid ambiguity (as noted, they must be *necessary, unambiguous, complete, consistent*).